complex model

2026-04-19

Model 2: Hierarchical varying-intercept and varying-slope model

Let

$$y_i = \log(\text{SalePrice}_i), \qquad i = 1, \dots, n.$$

We assume

$$y_i \sim \mathcal{N}(\mu_i, \sigma^2),$$

with

$$\mu_i = \beta_0 + \alpha_{j[i]} + \left(\beta_2 + \gamma_{j[i]}\right) \text{LogGrLivArea}_i + \beta_1 \text{OverallQual}_i + \beta_3 \text{GarageCars}_i + \beta_4 \text{TotalBsmtSF}_i + \beta_5 \text{YearSinceRemodel}_i,$$

where $j[i]$ is the neighborhood of house i.

The neighborhood-specific effects are modeled hierarchically as

$$\alpha_j \sim \mathcal{N}(0, \tau_\alpha^2), \qquad j = 1, \dots, J,$$

$$\gamma_j \sim \mathcal{N}(0, \tau_\gamma^2), \qquad j = 1, \dots, J.$$

Equivalently, we may write

$$\begin{pmatrix} \alpha_j \\ \gamma_j \end{pmatrix} \sim \mathcal{N}\left(\begin{pmatrix} 0 \\ 0 \end{pmatrix}, \Sigma \right),$$

where

$$\Sigma = \begin{pmatrix} \tau_\alpha^2 & \rho\tau_\alpha\tau_\gamma \\ \rho\tau_\alpha\tau_\gamma & \tau_\gamma^2 \end{pmatrix}.$$

The priors are

$$\beta_0 \sim \mathcal{N}(8.8, 0.05^2),$$
$$\beta_1 \sim \mathcal{N}(0.10, 0.03^2), \quad \beta_2 \sim \mathcal{N}(0.25, 0.08^2), \quad \beta_3 \sim \mathcal{N}(0.06, 0.03^2),$$
$$\beta_4 \sim \mathcal{N}(0.00010, 0.00003^2), \quad \beta_5 \sim \mathcal{N}(-0.003, 0.004^2),$$
$$\tau_\alpha \sim \text{half-Normal}(0, 0.12^2), \qquad \tau_\gamma \sim \text{half-Normal}(0, 0.05^2),$$
$$\rho \sim \text{Uniform}(-1, 1), \qquad \sigma \sim \text{half-Normal}(0, 0.10^2).$$